

KRP Proposal — Notation Table

Companion reference for *Research Proposal — Kinematic Recovery Perturbation (KRP)*. Symbols are grouped by role. Superscript 0 denotes the ego vehicle; superscript $i > 0$ denotes background vehicles.

1. State and perturbation (§1, §3.4)

Symbol	Meaning
s_t	State at time t (position x, y , heading ψ , speed v)
s_t^0	Ego state (superscript 0 = ego)
$s_t^{i>0}$	Background-vehicle state (superscript $i > 0$ = non-ego)
s_0	State at the current frame ($t = 0$) — note: subscript 0 = frame index, not ego
ξ	RAP's state offset perturbation, $\xi \sim \mathcal{U}(\Xi)$
Ξ	Sampling range of ξ
$\hat{\xi}$	Offset read off from frame-to-frame inconsistency (the shortcut), $\hat{\xi} \approx s_0 - \kappa(s_{-1}, a_{-1})$
$\Delta x, \Delta y$	Lateral / longitudinal position offset
$\Delta \psi$	Heading offset
Δv	Speed offset

2. Kinematics and physics (§1, §3.4)

Symbol	Meaning
κ	Differentiable kinematic bicycle model (state transition)
ω	Yaw rate
a_{lat}	Lateral acceleration
v	Speed
Δt	Inter-frame interval (= 0.5 s in NAVSIM)
δ_{steer}	Front-wheel steering angle
L	Wheelbase

3. Actions and optimization variables

Symbol	Meaning
a_t	Generic action fed to κ ; in E2 (KING) it is the free optimization variable
a_t^0	Ego action; $a_t^{i>0}$ background-vehicle action
$a_t^{\log}$	Ego's logged action from the dataset (a known constant)
$w_t, w_{-K:0}$	Ego control residual disturbance — the KRP optimization variable
w^*	The optimized (minimum-norm failure-inducing) disturbance
ε	Failure threshold; $\mathcal{L}_{\text{rec}} > \varepsilon$ marks "learner fails"
$\ \cdot\ _W$	Weighted norm; W is the weighting matrix
$\mathcal{W}(v_t)$	Speed-dependent actuation-limit set (calligraphic $\mathcal{W}$, distinct from the weight W)
K	History-window length (number of frames back)
$-K:0, -K:T$	Time-index ranges

Note (E3). The applied ego action is $a_t^{\log} + w_t = \text{logged nominal (fixed) + disturbance (optimized)}$. Unlike KING — which optimizes the full background action a_t — the ego already has a real logged trajectory, so KRP optimizes only the residual w_t . This is what makes the minimum-norm objective $\min_w \|w\|_W$ meaningful (smallest deviation from natural driving) and keeps perturbed clips near the data manifold. It follows the DART principle of injecting disturbances into the supervisor's control stream.

4. Cost, policy, and observation (§2, §3)

Symbol	Meaning
$\mathcal{C}$	KING's cost (geometric : collision distance)
$\mathcal{L}_{\text{rec}}$	KRP's recovery cost (perception-driven)
τ	Planner output trajectory
π_ω	Policy in KING (parameters ω)
π_θ	Learner / planner in KRP (parameters θ)
o_t	Observation (rendered image / BEV)
$\mathcal{R}$	Renderer / rasterizer
$\mathcal{R}_{\text{diff}}$	Differentiable rasterizer (modification M2; our contribution)
$\mathcal{M}$	Map (static scene)
x_{goal}	Goal location
$\rho(\theta)$	Directional radius of the recoverable region (evaluation metric)

5. Gradient factors (§3.2)

Symbol	Meaning
$\frac{\partial \mathcal{L}_{\text{rec}}}{\partial \tau}$	Cost w.r.t. trajectory
$\frac{\partial \pi_{\theta}}{\partial o}$	Planner w.r.t. observation (automatic differentiation)
$\frac{\partial o}{\partial s}$	Observation w.r.t. state — the sole bottleneck , available only under rasterization
$\frac{\partial s}{\partial w}$	State w.r.t. disturbance (bicycle model, closed form)
$T(\delta)$	Coordinate transform induced by pose offset δ ; first link of the $s \mapsto o$ chain

6. Index conventions

Convention	Meaning
Superscript 0	Ego vehicle (e.g. s_t^0, a_t^0)
Superscript $i > 0$	Background vehicle
Subscript t	Time step
Subscript 0	The current frame ($t = 0$)
log superscript	Quantity taken from the driving log

Overloaded symbols — read with care

Symbol	Meaning A	Meaning B
θ	In E2 / KING: the optimization variable $\{a_t^{i>0}\}$ (background actions)	In π_{θ} : the planner's network parameters
0	Superscript s^0 : ego vehicle	Subscript s_0 : frame at $t = 0$
W	Weight matrix in the norm $\ w\ _W$	Calligraphic $\mathcal{W}(v_t)$: actuation-limit set

Recommendation: in the KING paragraph, avoid introducing θ for the optimization variable — write $\{a_t^{i>0}\}$ directly — so that θ unambiguously denotes the planner parameters π_{θ} throughout.